

**K S Tharunpranav 251801359-2025-AIDS** ▾**K2****Started on** Wednesday, 16 September 2026, 5:46 PM**State** Finished**Completed on** Wednesday, 16 September 2026, 6:24 PM**Time taken** 37 mins 53 secs**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 10.00 out of 10.00**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3. Write any efficient algorithm to find the possible ways.

Example 1:**Input:** 6**Output:** 6**Explanation:** There are 6 ways to 6 represent number with 1 and 3

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format**Print: The number of possible ways 'n' can be represented using 1 and 3**

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      if (scanf("%d", &n) != 1) return 0;
6
7      unsigned long long dp[n + 1];
8
9      dp[0] = 1;
10     if (n >= 1) dp[1] = 1;
11     if (n >= 2) dp[2] = 1;
12
13     for (int i = 3; i <= n; i++) {
14         dp[i] = dp[i - 1] + dp[i - 3];
15     }
16
17     printf("%llu\n", dp[n]);
18
19     return 0;
20 }
```

	Input	Expected	Got	
✓	6	6	6	✓

	Input	Expected	Got	
✓	25	8641	8641	✓
✓	100	24382819596721629	24382819596721629	✓

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

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